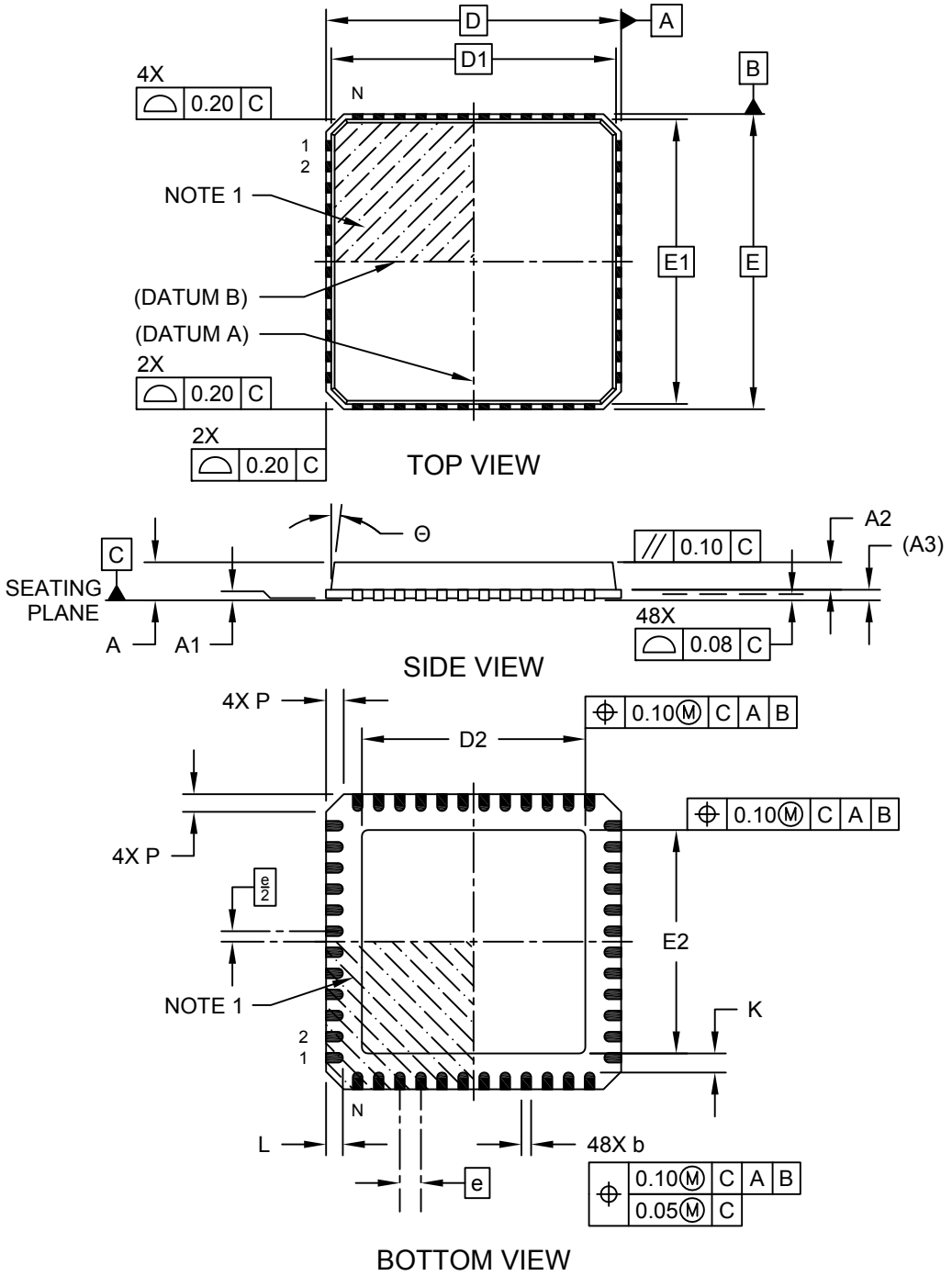
48-Lead Very Thin Plastic Quad Flat, No Lead Package (VQ) - 7x7 mm Body [VQFN] With 5.3 mm Exposed Pad; Punch Singulated

Note: For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



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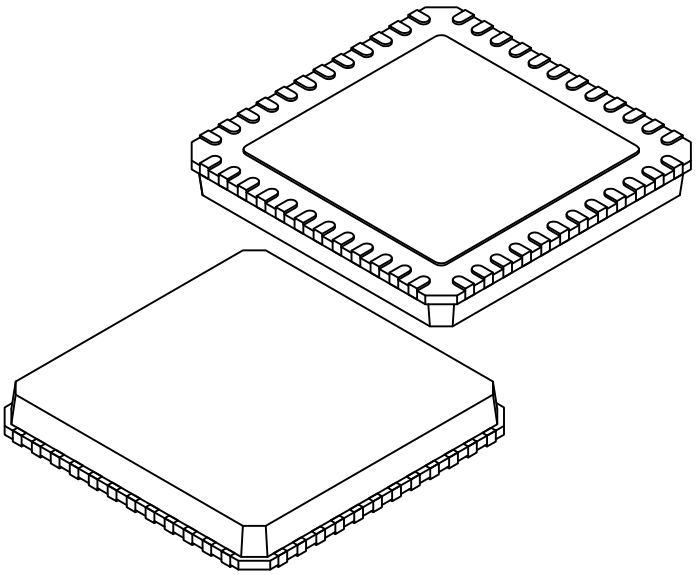

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	48		
Pitch	e	0.50 BSC		
Overall Height	A	0.80	0.85	0.90
Standoff	A1	0.00	0.02	0.05
Mold Cap Thickness	A2	0.60	0.65	0.70
Terminal Thickness	(A3)	0.20 REF		
Overall Length	D	7.00 BSC		
Mold Cap Length	D1	6.75 BSC		
Exposed Pad Length	D2	5.20	5.30	5.40
Overall Width	E	7.00 BSC		
Mold Cap Width	E1	6.75 BSC		
Exposed Pad Width	E2	5.20	5.30	5.40
Corner Chamfer	P	0.24	0.42	0.60
Terminal Width	b	0.18	0.23	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	0.20	-	-
Mold Cap Draft Angle	Θ	0°	-	14°

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is punch singulated
3. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

   REF: Reference Dimension, usually without tolerance, for information purposes only.

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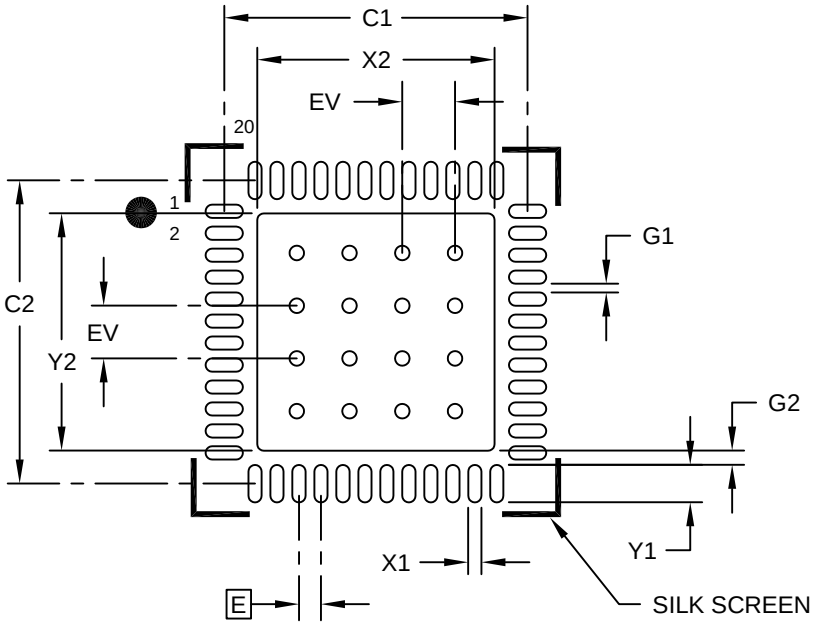
Note: For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Optional Center Pad Width	X2			5.40
Optional Center Pad Length	Y2			5.40
Contact Pad Spacing	C1		6.90	
Contact Pad Spacing	C2		6.90	
Contact Pad Width (X48)	X1			0.30
Contact Pad Length (X48)	Y1			0.85
Space Between Pads (X44)	G1	0.20		
Contact Pad to Center Pad (X48)	G2	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process